Zero Trust Networking with Tungsten Fabric

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Submitted by:

Group 5

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Abstract

Zero Trust Networking (ZT) represents a fundamental shift in cybersecurity strategy. Traditional perimeter-based security models, which assume everything inside the network is trusted, are no longer effective against sophisticated cyberattacks, insider threats, and unauthorized lateral movement. Traditional security relies heavily on established perimeter defenses, trusting users and devices once they cross network boundaries. This growing complexity has created a critical need for a more adaptive and resilient approach to network security. To address this challenge, software-defined networking (SDN) technologies play a vital role by decoupling network management from underlying hardware. Tungsten Fabric (TF) is an open-source SDN controller that provides network virtualization, policy-based traffic management, analytics, and integration with cloud environments such as Kubernetes and OpenStack. This report explores its key principles, implementation challenges, and how TUNGsten Fabric supports Zero Trust deployment by enabling dynamic, fine-grained network security policies. This study highlights the importance of integrating Zero Trust principles with SDN technologies to enhance network visibility, automate security policy enforcement, and reduce vulnerabilities in dynamic IT ecosystems.

1 Introduction

1.1 Background

In today's cloud-driven and virtualized environments, traditional perimeter-based network security is no longer effective against modern cyber threats. As organizations expand across cloud, IoT, and remote work infrastructures, network boundaries have become increasingly blurred. The Zero Trust Networking (ZTN) model, based on the principle of "never trust, always verify," ensures that every entity—user, device, or application—is continuously authenticated and authorized before accessing resources [1, 2, 6, 7]. This shift marks a fundamental move from implicit to adaptive, context-aware trust.

1.2 Problem Statement

Even with its increasing significance, putting Zero Trust into practice in actual networks is still a difficult task. Organizations often struggle to integrate ZTN principles with legacy systems and enforce granular, context-based policies across distributed environments. A recent 2025 review found that scalability, interoperability, and policy automation are major barriers to full Zero Trust adoption [5, 23]. Therefore, to show how Zero Trust can be implemented successfully in virtualized network environments, workable, open-source solutions are required.

1.3 Motivation and Objective

The motivation behind this project stems from the need to operationalize Zero Trust concepts using Software Defined Networking (SDN) technologies, which enable centralized, programmable control of network resources [4, 20]. Tungsten Fabric (TF), an open-source SDN platform, offers network virtualization, micro-segmentation, and analytics features that align closely with Zero Trust principles [3, 8, 9].

The project titled *"Implementation of Zero Trust Networking Principles using Tungsten Fabric"* aims to build a secure, policy-driven network that enforces identity-based access control and minimizes lateral movement between virtual machines.

1.4 Scope and Approach

The project establishes a virtualized lab environment with separate network zones for application and database layers. Using TF's centralized control plane, fine-grained access policies are applied to ensure that only authorized communication occurs between zones. Continuous monitoring and analytics visualize traffic flows and validate Zero Trust enforcement. This practical implementation demonstrates how SDN can automate and strengthen Zero Trust security through micro-segmentation and policy-driven management [1, 3, 4, 24].

1.5 Term Description

- **Zero Trust Networking (ZTN):** A model that continuously confirms the identities of users and devices before allowing access[1, 2, 6].
- **Tungsten Fabric (TF):** An open-source SDN platform enabling policy-based traffic management and network automation [3, 8].
- **Software Defined Networking (SDN):** A network architecture that decouples control and data planes for centralized management [4, 20].
- **Micro-segmentation:** Division of networks into isolated segments to prevent unauthorized movement [1, 23].

2 Literature Review / Background

The section that follows provides a thorough analysis of the literature on Tungsten Fabric, Software-Defined Networking, and Zero Trust Networking, emphasizing the conceptual underpinnings and interdependencies of each.

2.1 Understanding Zero Trust Networking

A major change in cybersecurity strategy is represented by Zero Trust Networking (ZT). Traditional security places a lot of trust in users and devices once they cross network boundaries and mainly depends on well-established perimeter defenses. However, this approach has become less effective with the rise of cloud computing, remote work, and mobile devices, which decentralize the traditional network perimeter. Zero Trust flips this model by assuming no entity—whether inside or outside the network—is inherently trustworthy. It requires continuous verification of every user, device, and application requesting access, minimizing implicit trust to reduce potential attack surfaces [1], [2].

This continuous trust assessment involves comprehensive identity verification, behavior monitoring, and strict access controls based on a "never trust, always verify" principle [5], [10]. Zero Trust emphasizes several key principles: least privilege access, micro-segmentation, multi-factor authentication, encryption of sensitive data, and real-time policy enforcement [18], [23]. Micro-segmentation plays a crucial role by dividing the network into smaller, isolated segments, limiting lateral movement opportunities for attackers even if an initial breach occurs. This approach significantly improves security posture in hybrid and multi-cloud environments where traditional perimeter boundaries are blurred [19], [20].

2.2 Zero Trust Models and Classification

Several Zero Trust models exist, varying in architecture and focus areas. Some models prioritize identity-centric approaches using robust authentication and authorization frameworks, while others emphasize network-centric segmentation and encryption [4], [6]. Comprehensive frameworks integrate identity, device, network, and application security into unified structures that dynamically adjust access rights and policies based on contextual risk parameters [7], [17]. Popular architectures include Google's BeyondCorp and NIST's Zero Trust Architecture, each highlighting continuous verification, device health checks, and contextual access determination [2], [18]. These models share a core goal: to minimize trust assumptions and enforce strict access limitations dynamically.

2.3 Real-World Applications and Challenges

Zero Trust is being adopted extensively across sectors like finance, healthcare, government, and critical infrastructure where sensitive data protection is paramount [13], [16]. For instance, healthcare institutions adopt Zero Trust to secure electronic health records and medical devices in complex environments. Despite its advantages, transitioning to Zero Trust is challenging. These include resistance to cultural shifts, integration complexity with legacy systems, perfor-

mance overhead due to constant verification, and balancing stringent security with user convenience [14], [15]. Additionally, maintaining privacy while monitoring is a sensitive issue. Organizations often incrementally adopt Zero Trust, focusing first on high-value assets and expanding iteratively [18], [23].

2.4 The Role of Software-Defined Networking

Software-Defined Networking (SDN) is a key enabler for Zero Trust networking. SDN decouples the control plane (decision-making) from the data plane (traffic forwarding), enabling programmable and centralized network management [20]. This separation allows for dynamic, fine-grained network segmentation and policy enforcement aligned with Zero Trust principles. SDN controllers manage network behavior based on real-time organizational policies, making them effective in implementing micro-segmentation, traffic filtering, and dynamic threat mitigation [21], [19].

2.5 Tungsten Fabric: Features and Relevance

Tungsten Fabric is an open-source SDN controller widely used in virtualized and containerized network environments [3], [4], [8]. It supports multi-tenancy, service chaining, and detailed analytics, which are essential for enforcing granular Zero Trust policies. Tungsten Fabric integrates well with Kubernetes and other cloud-native platforms, making it suitable for orchestrating complex network topologies in edge, hybrid, and multi-cloud deployments [7]. In addition to supporting automation and AI-driven analytics for proactive threat detection, it offers centralized management and visibility, allowing security teams to dynamically monitor and control access. [5], [12].

2.6 Synergy of Zero Trust and Tungsten Fabric

Combining Zero Trust with Tungsten Fabric offers a robust framework for securing modern distributed environments [13], [15]. Tungsten Fabric's programmability enables enforcement points to adapt policies based on real-time context and telemetry, embodying Zero Trust's core principle of continuous verification [18], [19]. Its integration with container orchestration platforms supports securing microservices and ephemeral workloads, which traditional networks find challenging to protect. This synergy enhances adaptive security, improves threat response times, and enables scalable segmentation across hybrid infrastructures [8], [9]. Challenges such as orchestrating policy consistency, ensuring low-latency enforcement, and managing heterogeneous resources require further research [25].

2.7 Related Concepts: SASE and NFV

Zero Trust also intersects with concepts like Secure Access Service Edge (SASE) and Network Function Virtualization (NFV) [6], [7], [20]. SASE blends network security functions with WAN capabilities, delivering secure access from cloud edges to users regardless of location. Zero Trust principles naturally align with SASE's focus on identity-centric, context-aware access. NFV complements SDN by virtualizing network functions, such as firewalls and intrusion detection, enabling flexible deployment and integration of security controls crucial for Zero Trust implementations [21], [23].

2.8 Emerging Technologies and Future Directions

The future of Zero Trust networking lies in integrating emerging technologies like artificial intelligence, machine learning, blockchain, and post-quantum cryptography [22], [24]. AI and ML improve anomaly detection, automate threat response, and optimize trust evaluation algorithms, while blockchain offers decentralized trust models. Post-quantum cryptography promises resilient encryption against future quantum threats [25]. Tungsten Fabric's extensible architecture makes it an ideal platform to incorporate these innovations, particularly in managing edge and hybrid cloud networks where agility and security must co-exist.

Researchers continue to explore enhanced trust evaluation, orchestration of distributed trust, and user-centric privacy-preserving mechanisms within Zero Trust frameworks. Standards development and best practices are also evolving to help organizations adopt Zero Trust effectively [16], [19], [23].

3 Methodology

Explain your research or implementation methods here — setup, tools, network configuration, simulation environment, etc.

4 Results

Present and discuss your main results, data, figures, or analysis.

5 Conclusion

Summarize your findings, lessons learned, and possible improvements. Discuss how Zero Trust with Tungsten Fabric can shape secure network infrastructures in the future.

6 References

References

[1] Y. Liu et al., "Hierarchical Micro-Segmentations for Zero-Trust Services via Large Language Model-enhanced Graph Diffusion," *arXiv preprint arXiv:2406.13964*, Jun. 2024.

[2] S. Raj Pokhrel, L. Yang, S. Rajasegarar, and G. Li, "Robust Zero Trust Architecture: Joint Blockchain-based Federated Learning and Anomaly Detection Framework," *arXiv preprint arXiv:2406.17172*, Jun. 2024.

[3] S. Arora and J. Hastings, "Microsegmented Cloud Network Architecture Using Open-Source Tools for a Zero Trust Foundation," *arXiv preprint arXiv:2411.12162*, Nov. 2024.

[4] J. A. Rahim et al., "Open-Source Software Defined Networking Controllers," *Journal of Network and Computer Applications*, 2024.

[5] M. D. N. Rahman et al., "Zero Trust Architecture: A Systematic Literature Review," *arXiv preprint arXiv:2503.11659*, 2025.

[6] J. Kindervag, "No More Chewy Centers: The Zero Trust Model of Information Security," *Forrester Research*, 2010.

[7] S. Rose, O. Borchert, S. Mitchell, and S. Connelly, *Zero Trust Architecture*, NIST Special Publication 800-207, National Institute of Standards and Technology, 2020.

[8] "Tungsten Fabric Documentation," Tungsten Fabric Project. Available: `https://tungsten.io/`

[9] S. Krasuski, "Tungsten Fabric Architecture — An Overview," *CodiLime Blog*, Jan. 2021. Available: `https://codilime.com/blog/tungsten-fabric-architecture-an-overview/`

[10] Intel Network Builders, "Zero Trust Reference Architecture Technology Guide," Nov. 2022. Available: `https://builders.intel.com/solutionslibrary/zero-trust-zero-trust-reference-architecture-technology-guide`

[11] U.S. Department of Defense, *Zero Trust Reference Architecture (V2.0)*, Jul. 2022. Available: `https://dodcio.defense.gov/`

[12] Linux Foundation Networking (LFN), *LF Networking White Paper*, Jun. 2022. Available: `https://lfnetworking.org`

[13] Y. Yang, "Tungsten Fabric Overview and Intel Contributions," *Intel and Linux Foundation*, 2018.

[14] A. Arora and J. Hastings, "Micro-segmented Cloud Network Architecture Using Open-Source Tools for a Zero Trust Foundation," *arXiv preprint arXiv:2411.12162*, 2024.

[15] Palo Alto Networks, "What Is a Zero Trust Architecture (ZTA)?," Oct. 2022. Available: `https://www.paloaltonetworks.com/cyberpedia/what-is-a-zero-trust-architecture`

[16] TechTarget, "What Is the Zero-Trust Security Model?," Oct. 2022. Available: `https://www.techtarget.com/searchsecurity/definition/zero-trust-model-zero-trust-network`

[17] Mobile World Live, "Zero Trust Starts Here: Why Strong, Future-Proof Encryption Is the Bedrock of Modern Cybersecurity," Apr. 2025.

[18] P. Dhiman, N. Saini, Y. Gulzar, S. Turaev, A. Kaur, K. U. Nisa, and Y. Hamid, "A Review and Comparative Analysis of Relevant Approaches of Zero Trust Network Model," *Sensors*, vol. 24, no. 4, p. 1328, Feb. 2024, doi: 10.3390/s24041328.

[19] S. Ghasemshirazi, G. Shirvani, and M. A. Alipour, "Zero Trust Applications, Challenges, and Opportunities," [Online]. Available: attached file "ZT.pdf," 2025.

[20] B. U. Kazi, M. K. Islam, M. M. H. Siddiqui, and M. Jaseemuddin, "A Survey on Software Defined Network-Enabled Edge Cloud Networks: Challenges and Future Research Directions," *Network*, vol. 5, no. 2, p. 16, May 2025, doi: 10.3390/network5020016.

[21] J. Kindervag, "Build Security into Your Network's DNA: The Zero Trust Network Architecture," *Forrester Research*, 2010.

[22] NIST Special Publication 800-207, "Zero Trust Architecture," Aug. 2020.

[23] C. Buck, C. Olenberger, A. Schweizer, F. Völter, and T. Eymann, "Never Trust, Always Verify: A Multivocal Literature Review on Current Knowledge and Research Gaps of Zero-Trust," *Computer Security*, vol. 110, p. 102436, 2021.

[24] P. Assunção, et al., "A Zero Trust Architecture for Cloud Computing," *IEEE Transactions on Cloud Computing*, vol. 9, no. 1, pp. 1–8, 2021.

[25] M. Mao, et al., "Mobile Edge Computing: Survey and Research Outlook," *IEEE Communications Magazine*, vol. 54, no. 4, pp. 36–42, Apr. 2016.

[26] “Open Network Foundation,” “OpenFlow Switch Specification,” Version 1.5.1, Dec. 2015.

[27] Z. Chen, et al., “Research on Zero-Trust Security Protection Technology of Power IoT based on Blockchain,” *Journal of Physics: Conference Series*, vol. 1769, 2021.

[28] S. Syed, et al., “Zero Trust Architecture (ZTA): A Comprehensive Survey,” *IEEE Access*, vol. 10, pp. 714379–714400, 2022.

[29] P. Velumani, et al., “Secured Secret Sharing of QR Codes Based on Nonnegative Matrix Factorization and Regularized Super Resolution Convolutional Neural Network,” *Sensors*, vol. 22, 2022.

[30] B. U. Kazi, et al., “Resource Allocation Frameworks for SDN-Enabled Edge Cloud,” *Network*, 2025.